

Top-down control or Bottom-up release: Evidence for predation‘s grip on prey numbers

Species Interactions & Coexistence

Vaghadia Saket Dilip, Singh Angleen,
Mejsnarová Veronika, Kerkhoff Sigrid

Ecological Concepts

TOP-DOWN CONTROL

- Predator pressure directly regulates prey abundance and limits population growth

BOTTOM-UP CONTROL

- Food availability, food quality, and primary plant productivity dictate herbivore carrying capacity

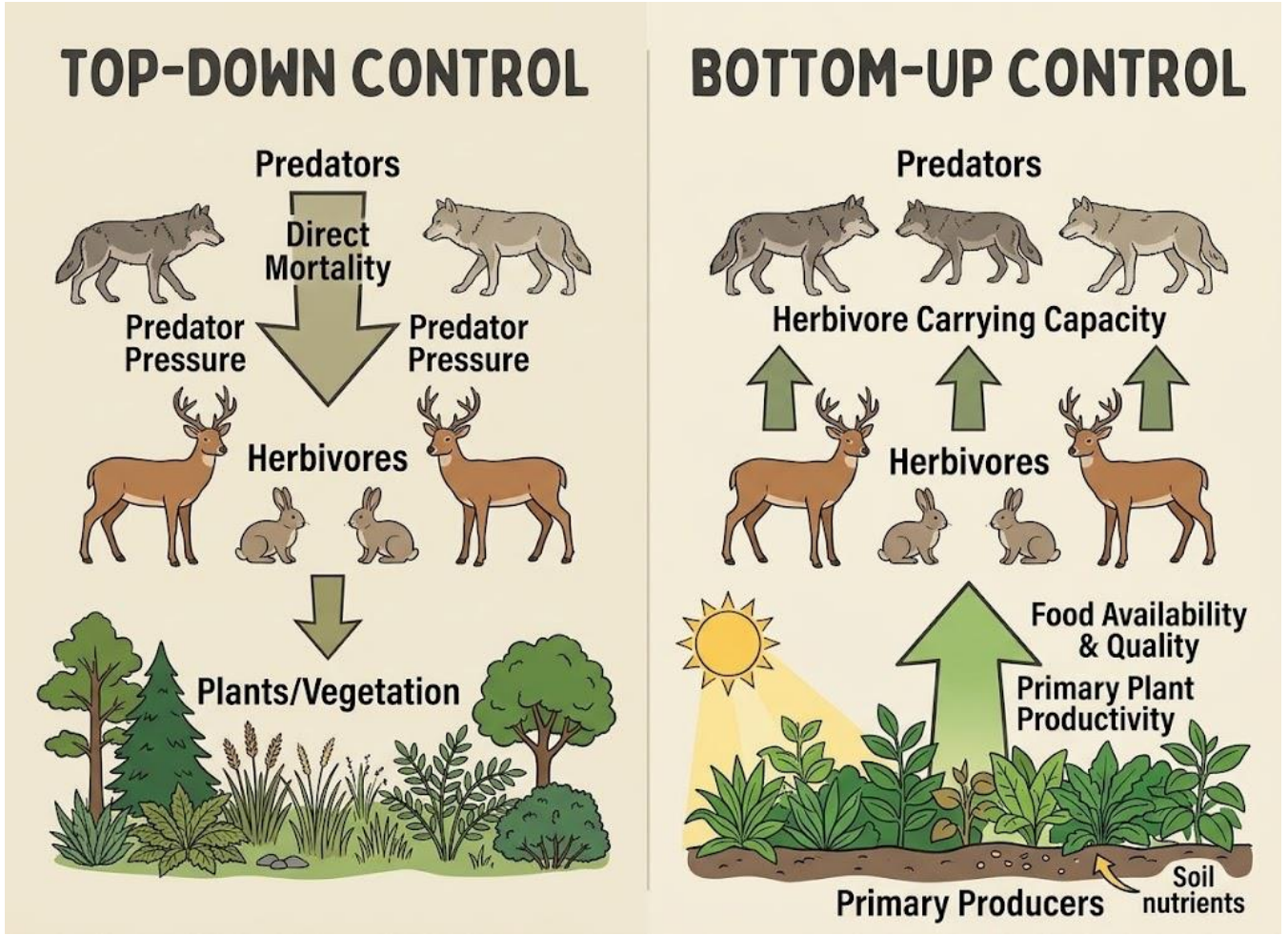


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Ecological Concepts

PREDATION-PREY INTERACTIONS

LETHAL EFFECTS

Direct consumption and killing of prey

NON-LETHAL EFFECTS

Risk-induced stress and behavioral shifts

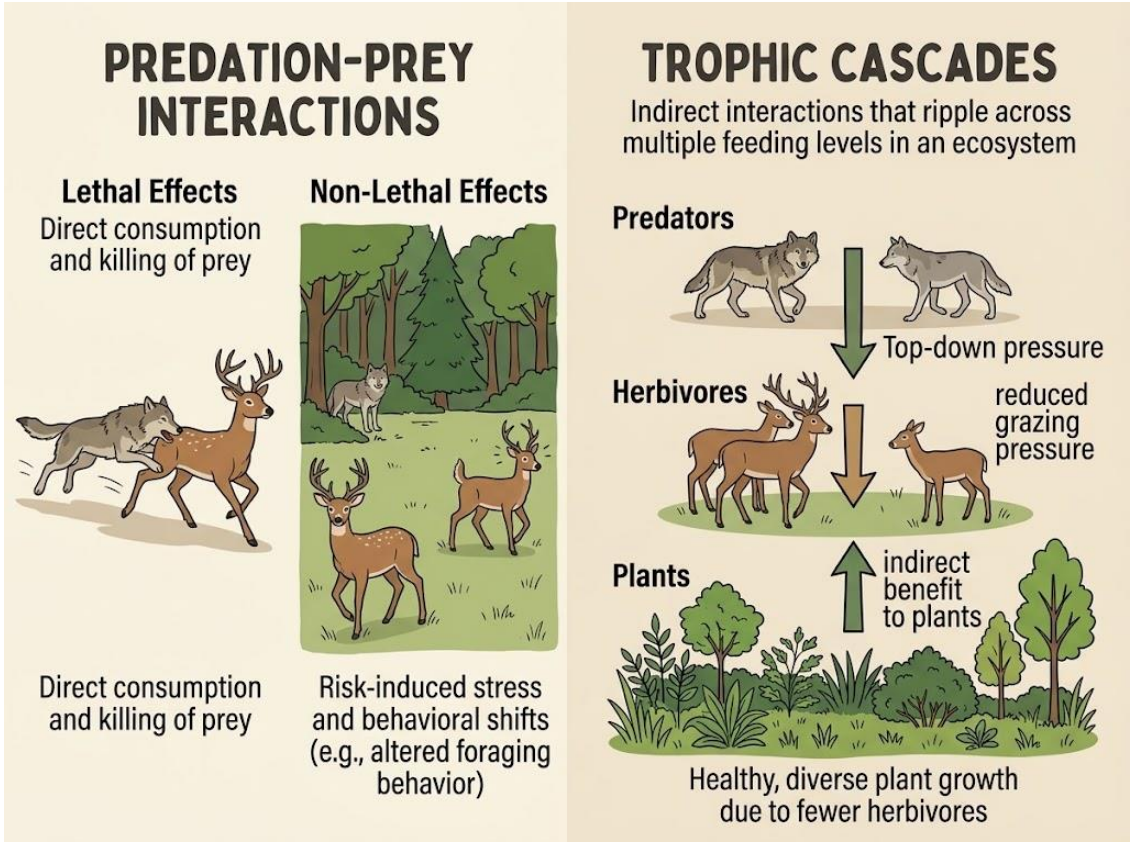


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Trophic Cascades

Indirect interactions that ripple across multiple feeding levels in an ecosystem

Study System Overview

➔ Snowshoe hare (Lepus americanus) in the boreal forest of Kluane, Yukon, Canada

➔ A regular 9-to-11-year population cycle where hare numbers grow very large and then sharply crash

Three-Trophic-Level System = **THE FOOD WEB**

LEVEL 1 – **PLANTS**
● Primary Producers

LEVEL 2 – **HARES**
● Primary Consumers

LEVEL 3 – **PREDATORS**
● Secondary Consumers

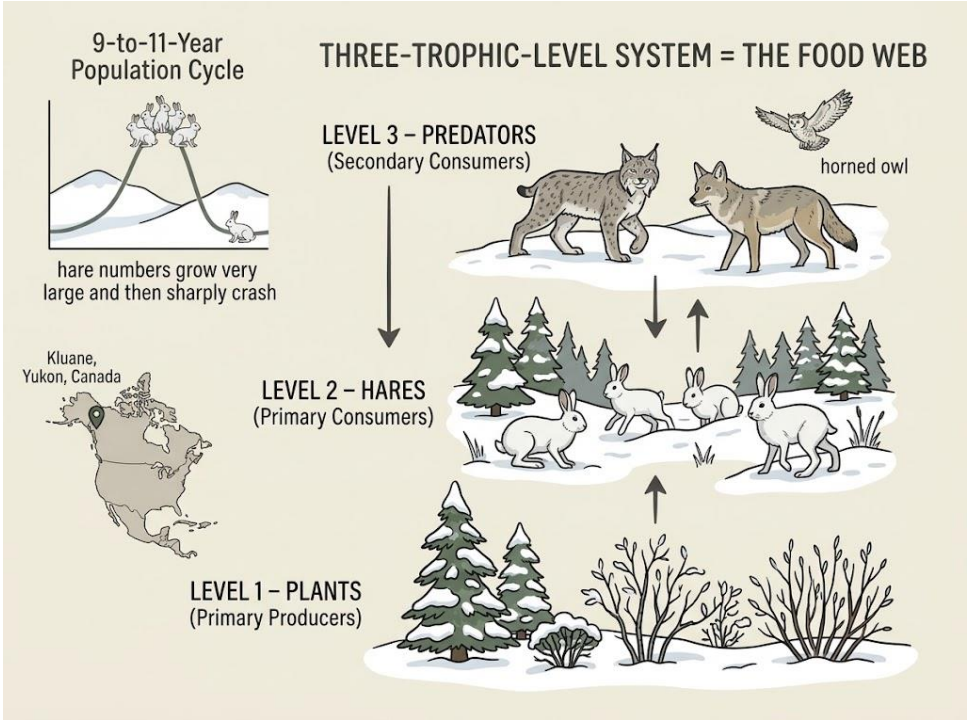


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PAPER 1 | 1995

Impact of Food and Predation on the Snowshoe Hare Cycle

Charles J. Krebs and colleagues

Science, 269 (5227), 1112–1115 (1995)

Purpose:

To investigate how food availability and predation influence snowshoe hare populations.

Study location:

Kluane Lake region, Yukon, Canada

How Did the Researchers Test it?

Researchers compared forest plots with different experimental treatments.

1. Control
No extra food; predators had access.

2. Food addition
Extra food was provided to hares.

3. Predator exclusion
Fences reduced access by mammalian predators.

4. Combined treatment
Extra food and predator exclusion together.

Additional test: Forest Fertilization
Separate plots were fertilized to test whether increasing plant growth affected hare density.





What Were the Main Results?

Approximate hare-density differences reported during the peak and decline phases.



Important result: The combined effect was greater than either treatment alone

Source: Krebs et al. (1995), Science, 269(5227), 1112–1115.

What Do The Results Tell Us?

Krebs et al. (1995) — Interpretation

Top-Down Control
Predator exclusion increased hare density, supporting an effect of predation.

Bottom-Up Effects
Adding food increased hare density, showing food availability also matters.

Interaction
Food + predator exclusion produced the largest increase (~11×).

Conclusion: neither food nor predation alone explains the full cycle.

PAPER 2 | 2026

Demographic mechanisms of snowshoe hare population cycles in Yukon, Canada.

Madan K. Oli and colleagues.

- Research question: which demographic processes drive population growth and decline?
- Study area: Kluane Lake, Yukon, Canada.



Steve Sayles from Rankin Inlet, Nunavut, Canada - Arctic_Hare

Methods: how did they investigate?

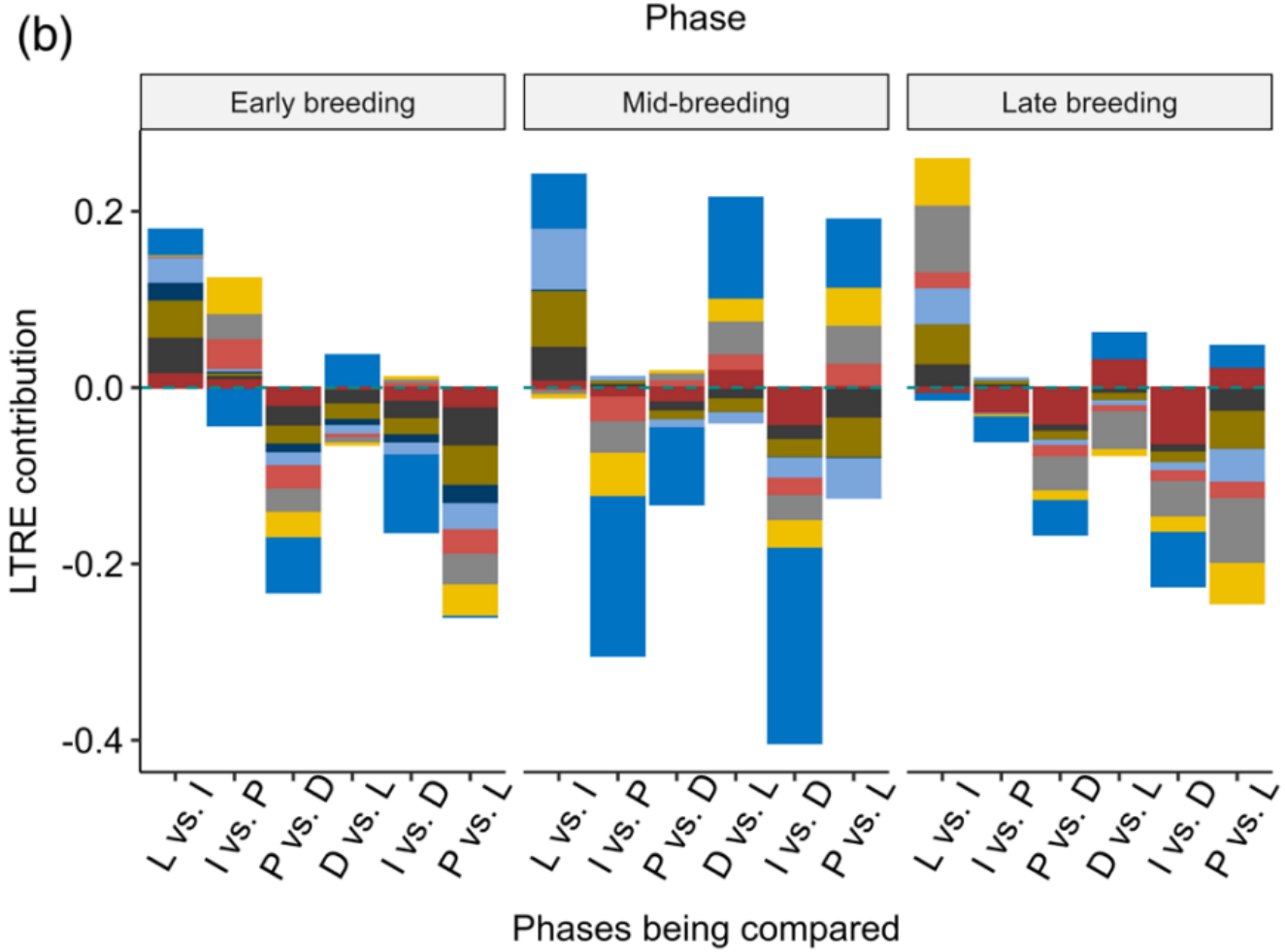
- Data from 1977-2020.
- Snowshoe hare population cycles 9-11 years
- More than 20,000 captures (more than 7,000 individual hares).
- Method: capture-mark-recapture models
- Population growth, survival and reproduction analysed across cycle phases.



Image made by AI

Results

- Main findings:
- Survival = higher during population increase phase.
- Population decline because of reduced survival, mainly before weaning.
- Dark blue: pre-weaning survival, yellow: juveniles survival
- Population growth and decline. Olive green and black: females (becoming) reproductive
- Adult female survival



Oli, M. K., Kenney, A. J., Boonstra, R., Boutin, S., Murray, D. L., Jung, T. S., Hines, J. E., & Krebs, C. J. (2026). *Demographic mechanisms of snowshoe hare population cycles in Yukon, Canada*. *Journal of Animal Ecology*, 95, 21–38. https://doi.org/10.1111/1365-2656.70169

What do the results mean for top-down and bottom-up control?

- Predation
- Survival, phase of the cycle.
- Reproduction and pre-weaning survival
- The 1995 experiment and 2026 demographic study provide complementary evidence.

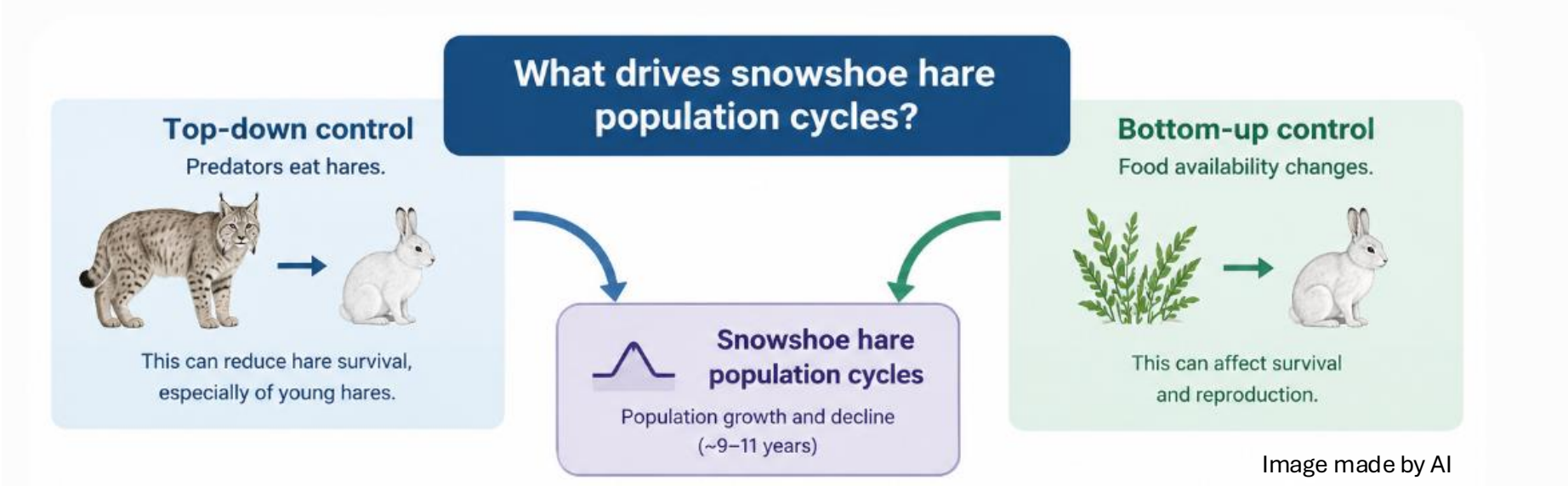
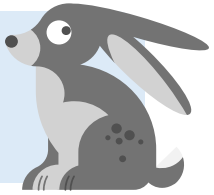


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Two Studies, Two Kinds of Evidence



- 1995 · Krebs et al.
- EXPERIMENTAL EVIDENCE
- Method: Food addition + predator exclusion
- Finding: Strong interaction between both factors
- Limitation: Population decline was delayed, not prevented

- 2026 · Oli et al.
- DEMOGRAPHIC EVIDENCE
- Method: 43 years of population data
- Finding: Survival and reproduction drive population changes
- Limitation: Ecological causes not directly tested

1995 tested ecological pressures; 2026 explained the demographic processes behind population changes.

Sources: Krebs et al. (1995); Oli et al. (2026).

Why Didn't the Population Decline Stop?

KREBS ET AL. · 1995

- **The unexpected result**
- Food + predator exclusion increased hare density
- Population still eventually declined
- Birds of prey remained, and hares could move outside protected areas.

OLI ET AL. · 2026

- **A deeper insight**
- Survival and reproduction shape population cycles
- Pre-weaning survival is crucial during declines
- Causes of reproductive changes remain unclear

The bigger picture

Increasing hare numbers does not necessarily mean the natural population cycle has been eliminated or fully explained.

Sources: Krebs et al. (1995); Oli et al. (2026).

Final Verdict: What Have We Learned?

What the evidence supports

- Predation strongly influences hare survival
- Food and predation interact
- Survival and reproduction shape population cycles

What remains uncertain

- Causes of reproductive changes.
- Why do hare populations remain low for several years?
- Do similar demographic patterns occur in other cyclic populations?

THE FINAL ANSWER

Predation plays a major role in controlling hare populations, but food availability and demographic processes are also essential to understanding the cycle.

Sources: Krebs et al. (1995); Oli et al. (2026).

Thank You
for your attention!

Any Questions?

